

Backend Engineering — Internship Assignment

AI-Powered Job Matching Platform

Role	Backend Intern
Duration	5–7 Days
Stack	Python · FastAPI · SQLite · Claude API

About the Project

We are building a job matching platform that helps Indian job seekers find the most relevant openings across major companies — without them having to manually browse dozens of career portals. A candidate fills out a Google Form, uploads their CV, and within minutes receives an email with their **top 5 job matches**, each with a compatibility score, skill gap analysis, and a direct apply link.

The backend pipeline works in three stages:

- 1. CV Parsing:** Claude API reads the raw CV text and extracts structured data — skills, experience, past roles, education.
- 2. Pre-filtering:** Simple skills-overlap math narrows ~3,600 live job listings down to ~50 plausible candidates. No LLM needed here — speed and cost matter.
- 3. Deep Matching:** The top 50 jobs are sent to Claude in batches of 10. Claude acts as a recruiting expert and scores each job 0–100 with a short reasoning note.

Your Task

Build a **minimal but working version** of the matching pipeline as a Python script or FastAPI endpoint. You are given sample data below — you do not need to scrape anything or build a UI.

Task 1: CV Parsing via Claude API

Given the sample CV text below, write a function that calls the Claude API and returns a structured JSON profile with: `candidate_name`, `skills` (list), `years_of_experience`, `preferred_locations` (list), and a 2-sentence summary.

Task 2: Pre-filter Jobs

Given the 5 sample job listings (provided below), write a function that scores each job purely on skills overlap with the candidate profile. Return jobs sorted by overlap score (highest first). No LLM needed for this step.

Task 3: Deep Match via Claude API

Take the top 3 pre-filtered jobs and send them to Claude in a single API call (batch of 3). Ask Claude to return a score (0–100) and a 1-sentence reason for each. Parse the response into structured JSON.

Task 4: Output

Print or return the final top 3 results in a clean format showing: rank, job title, company, match score, reason, and apply URL.

Sample Input Data

Sample CV Text (pass this as a string to your CV parser):

```
Name: Priya Sharma
Email: priya.sharma@email.com | Phone: +91-9876543210
Location: Bangalore

SUMMARY
Software engineer with 2 years of experience in Python backend development.
Passionate about building data pipelines and REST APIs.

SKILLS
Python, FastAPI, Django, PostgreSQL, REST APIs, Git,
Docker, basic ML concepts, Pandas, NumPy

EXPERIENCE
Junior Backend Developer – TechStart Pvt. Ltd. (2022–2024)
- Built REST APIs using FastAPI serving 50k+ daily requests
- Optimised PostgreSQL queries reducing load time by 40%
- Integrated third-party APIs (payment, SMS, email)

EDUCATION
B.Tech Computer Science – VTU, 2022 (CGPA: 8.4)
```

Sample Job Listings (use this list as your job database):

```
[
  {
    "id": "JOB001",
    "title": "Backend Engineer - Python",
```

```
    "company": "Infosys",
    "location": "Bangalore",
    "work_mode": "hybrid",
    "seniority": "junior",
    "required_skills": ["Python", "FastAPI", "PostgreSQL", "REST APIs", "Docker"],
    "description": "Build and maintain microservices using Python/FastAPI. Strong SQL skills required.",
    "apply_url": "https://careers.infosys.com/jobdescription/JOB001"
  },
  {
    "id": "JOB002",
    "title": "Data Engineer",
    "company": "TCS",
    "location": "Pune",
    "work_mode": "on-site",
    "seniority": "mid",
    "required_skills": ["Python", "Spark", "Hadoop", "SQL", "Airflow"],
    "description": "Design and manage large-scale data pipelines on cloud infrastructure.",
    "apply_url": "https://ibegin.tcs.com/jobs/JOB002"
  },
  {
    "id": "JOB003",
    "title": "ML Engineer",
    "company": "Google India",
    "location": "Hyderabad",
    "work_mode": "remote",
    "seniority": "senior",
    "required_skills": ["Python", "TensorFlow", "MLOps", "Kubernetes", "GCP"],
    "description": "Deploy and monitor production ML models at scale. 5+ years experience required.",
    "apply_url": "https://careers.google.com/jobs/JOB003"
  },
  {
    "id": "JOB004",
    "title": "Python Developer - APIs",
    "company": "Wipro",
    "location": "Bangalore",
    "work_mode": "hybrid",
    "seniority": "junior",
    "required_skills": ["Python", "Django", "REST APIs", "MySQL", "Git"],
    "description": "Develop RESTful APIs using Django. Work with cross-functional agile teams.",
    "apply_url": "https://careers.wipro.com/jobs/JOB004"
  },
  {
    "id": "JOB005",
    "title": "DevOps Engineer",
    "company": "Accenture",
    "location": "Chennai",
    "work_mode": "on-site",
    "seniority": "mid",
    "required_skills": ["Docker", "Kubernetes", "CI/CD", "AWS", "Terraform"],
    "description": "Manage cloud infrastructure and CI/CD pipelines for enterprise clients.",
    "apply_url": "https://www.accenture.com/in-en/careers/JOB005"
  }
]
```

Expected Output

Your script should produce output similar to the example below. The exact scores may vary — we care about the **structure and reasoning**, not specific numbers.

```
=== CV PARSE RESULT ===
{
  "candidate_name": "Priya Sharma",
  "skills": ["Python", "FastAPI", "Django", "PostgreSQL",
            "REST APIs", "Git", "Docker", "Pandas", "NumPy"],
  "years_of_experience": 2,
  "preferred_locations": ["Bangalore"],
  "summary": "Backend engineer with 2 years of Python and
             FastAPI experience. Strong in REST APIs and SQL
             optimisation."
}

=== PRE-FILTER SCORES (skills overlap) ===
JOB001 – Infosys Backend Engineer      overlap: 5/5 → 100%
JOB004 – Wipro Python Developer         overlap: 4/5 → 80%
JOB002 – TCS Data Engineer             overlap: 2/5 → 40%
JOB005 – Accenture DevOps Engineer     overlap: 1/5 → 20%
JOB003 – Google ML Engineer            overlap: 1/5 → 20%

=== DEEP MATCH (Claude API – top 3) ===

Rank 1 | Infosys – Backend Engineer – Python
Score : 91/100
Reason: Strong skills alignment – FastAPI, PostgreSQL, Docker,
       and REST APIs all match; junior seniority fits 2 YoE.
Apply : https://careers.infosys.com/jobdescription/JOB001

Rank 2 | Wipro – Python Developer – APIs
Score : 78/100
Reason: Good overlap on Python, Django, and REST APIs; MySQL vs
       PostgreSQL is a minor gap but easily bridgeable.
Apply : https://careers.wipro.com/jobs/JOB004

Rank 3 | TCS – Data Engineer
Score : 42/100
Reason: Python matches but candidate lacks Spark, Airflow, and
       Hadoop; also a mid-level role for a junior candidate.
Apply : https://ibegin.tcs.com/jobs/JOB002
```

Evaluation Criteria

Criterion	What we look for	Weight
Code Quality	Clean, readable, well-commented Python	25%
API Integration	Correct Claude API usage, prompt design, JSON parsing	30%

Logic Accuracy	Pre-filter overlap calculation, batch prompt design	25%
Output Structure	Final result matches expected format	10%
Bonus	FastAPI endpoint wrapping the pipeline	10%

What to Submit

1. A single Python script (or a small FastAPI app) with clear inline comments.
2. A README.md (even 5 lines) explaining how to run your code and your Claude API key setup.
3. Paste the actual terminal output of your script into the README under "Sample Run".
4. Push everything to a **public GitHub repo** and share the link.

■ *Tip: You can get a free Claude API key at console.anthropic.com. The Haiku model is cheapest for testing — budget under ■50 for this entire assignment.*

Questions? Reply to your Internshala message thread.

We review submissions on a rolling basis — quality over speed!